

FYM330 Problem set 3 assignment # 1

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1 Calculation of additional data from the input data

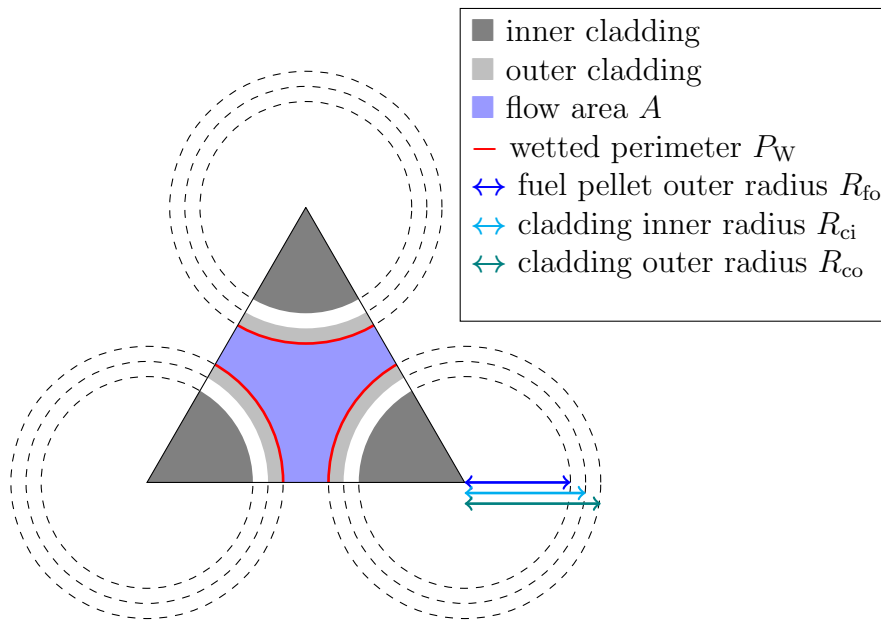


Figure 1: Geometry of the reactor. Our finite volume is the part with color.

In figure 1, we identify the wetted perimeter, marked in red, as half of the outer perimeter of a fuel rod. Concretely, we get

$$P_W = \frac{2\pi}{2} R_{co} = \pi R_{co}. \quad (1)$$

The flow area is the area of the entire triangle minus the area occupied by the cladding, which is half the area of an entire fuel rod. We get

$$A = \frac{\sqrt{3}}{4}p^2 - \frac{\pi}{2}R_{\text{co}}^2. \quad (2)$$

The area enclosed by a circle with perimeter s is $s^2/(4\pi)$. The equivalent hydraulic diameter thus solves the equation

$$\frac{(\pi D_e)^2/(4\pi)}{\pi D_e} = \frac{A}{P_W} \quad (3)$$

$$D_e/4 = \frac{A}{P_W} \quad (4)$$

$$D_e = \frac{4A}{P_W}. \quad (5)$$

The height DZ of the axial node solves $DZ \cdot NZ = H$, where H is the height of the channel. Hence

$$DZ = \frac{H}{NZ}. \quad (6)$$

The positions r_i of the NZ equisurface regions solve the difference equation

$$r_0 = 0 \quad (7)$$

$$r_{2NZ} = R_{\text{fo}} \quad (8)$$

$$r_{2NZ+2} = R_{\text{ci}} \quad (9)$$

$$r_{2NZ+4} = R_{\text{co}} \quad (10)$$

$$\pi r_i^2 - \pi r_{i-1}^2 = \pi r_{i+1}^2 - \pi r_i^2 \text{ for all other } i. \quad (11)$$

We rewrite the last equality to

$$r_i^2 = \frac{r_{i+1}^2 + r_{i-1}^2}{2} \quad (12)$$

and note that the sequence $(r_i^2)_{i=1}^{2NZ}$ is equispaced. Thus $r_i^2 = \frac{i}{2NZ}R_{\text{fo}}^2$, and so

$$r_i = \sqrt{\frac{i}{2NZ}}R_{\text{fo}} \quad (13)$$

$$r_{2NZ} = R_{\text{fo}} \quad (14)$$

$$r_{2NZ+1} = \sqrt{R_{\text{fo}}^2 + R_{\text{ci}}^2} \quad (15)$$

$$r_{2NZ+2} = R_{\text{ci}} \quad (16)$$

$$r_{2NZ+3} = \sqrt{R_{\text{ci}}^2 + R_{\text{co}}^2} \quad (17)$$

$$r_{2NZ+4} = R_{\text{co}}. \quad (18)$$

The corresponding perimeters are

$$S_i = 2\pi r_i, \tag{19}$$

and the corresponding surfaces are

$$V_1 = \pi r_1^2 \tag{20}$$

$$V_i = \pi r_i^2 - \pi r_{i-1}^2. \tag{21}$$